

2.3 Weather and Climate

Eduqas GCSE Geography B (C112QS) — Theme 2: Changing Environments

1. Weather vs Climate

Weather: The short-term atmospheric conditions at a specific place and time — including temperature, rainfall, wind speed, humidity, cloud cover, and air pressure. Weather can change from hour to hour.

Climate: The *average* weather conditions of a place over a **long period of time** (typically 30 years). Climate describes the expected pattern of weather for a region rather than day-to-day variation.

UK Climate Characteristics — Temperate Maritime

The UK has a **temperate maritime climate**, shaped primarily by its proximity to the Atlantic Ocean and the warming North Atlantic Drift current.

Season	Temperature	Rainfall	Key influence
Winter	Mild (rarely below −5°C in lowlands)	Wet; frequent frontal rain and depressions	Atlantic Ocean stores summer heat and releases it slowly; North Atlantic Drift
Summer	Warm (rarely above 28°C in most areas)	Drier, though convectional rain possible	Longer days increase solar radiation; Atlantic limits extremes
Autumn/Spring	Transitional — no sharp boundaries	Variable; changeable	Interaction of air masses

Key characteristics of the UK climate:

- No extreme temperatures — cool summers and mild winters (compared to same latitude inland)
- Rainfall distributed throughout the year (no dry season)
- Western areas receive more rainfall than eastern areas (prevailing south-westerly winds)
- Highly variable, changeable day-to-day weather
- Influenced by several different **air masses** (Polar Maritime, Tropical Maritime, Polar Continental, Tropical Continental)

Exam tip: Don't confuse weather and climate. Weather = short-term, specific. Climate = long-term, average. A common exam question asks you to “distinguish between weather and climate” — always include both the time-scale difference and the idea of averages vs day-to-day conditions.

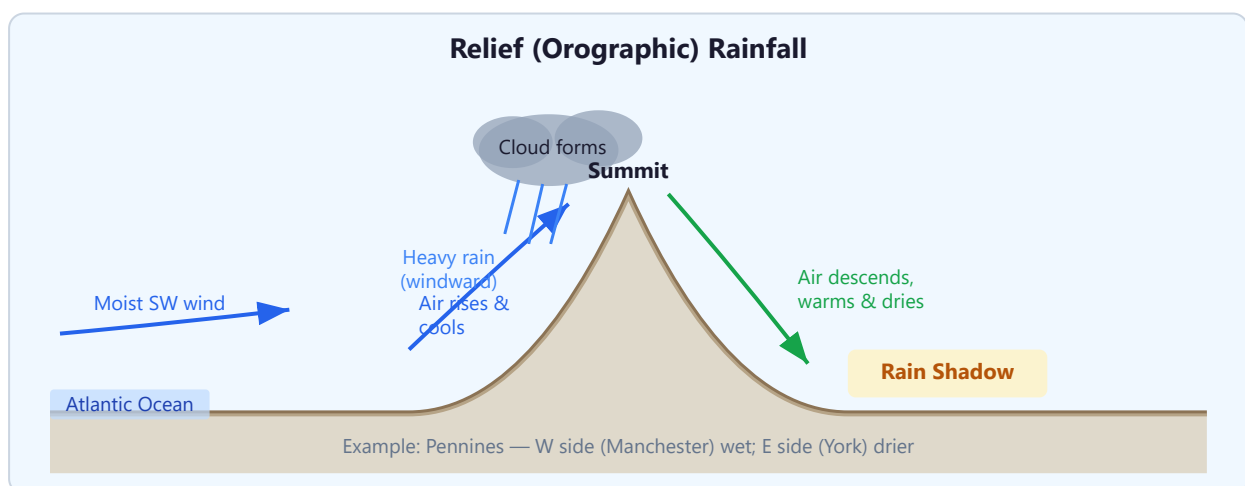
2. Factors Affecting UK Climate

Factor	Effect on UK Climate	Example
Latitude	The UK (50–61°N) receives less concentrated solar radiation than tropical regions because the Sun is lower in the sky. The same amount of solar energy is spread over a larger surface area. Higher latitudes are cooler.	Scotland is colder than southern England because it is at higher latitude (more northerly)
Altitude	Temperature decreases with height at an average rate of 6.5°C per 1,000 m (the environmental lapse rate). High ground is colder and receives more precipitation. Snow more likely on summits.	Cairngorm summit (1,245 m) is on average ~8°C colder than nearby Aviemore in the valley; over 100 days of snow cover/year
Distance from the sea (continentality)	The sea heats and cools more slowly than the land (<i>maritime effect</i>). Areas near the sea have smaller temperature ranges (mild winters, cool summers). Inland areas have a <i>continental climate</i> — hotter summers, colder winters, less rainfall.	London (inland) is hotter in summer and colder in winter than Plymouth (coastal). Eastern Europe has extremes not seen in W England.

Prevailing winds	UK's prevailing winds come from the south-west , bringing warm, moist air from the Atlantic Ocean. This gives the west coast more rainfall. Easterly winds from continental Europe bring cold, dry conditions in winter and hot, dry air in summer.	Cardiff receives c. 1,150 mm rain/year; London receives c. 590 mm/year (sheltered from south-westerly winds by Wales)
Ocean currents (North Atlantic Drift)	The North Atlantic Drift (NAD) — an extension of the Gulf Stream — flows north-eastward past the British Isles, bringing warm tropical water. This keeps the UK significantly warmer than other places at the same latitude (e.g. Newfoundland, Canada).	Stornoway, Scotland (58°N) has January average of 5°C; Labrador, Canada (same latitude) averages –15°C in January

3. Relief Rainfall

Relief (orographic) rainfall: Rainfall caused when moist air is forced to rise over a mountain barrier (orographic uplift). As the air rises, it cools (at the dry adiabatic lapse rate — 9.8°C/1,000 m until saturation; then at the saturated adiabatic lapse rate — ~6°C/1,000 m). Once cooled to the dew point, water vapour condenses into clouds and falls as rain on the *windward* side. On the *leeward* side (the rain shadow), descending air warms and dries out, producing lower rainfall.

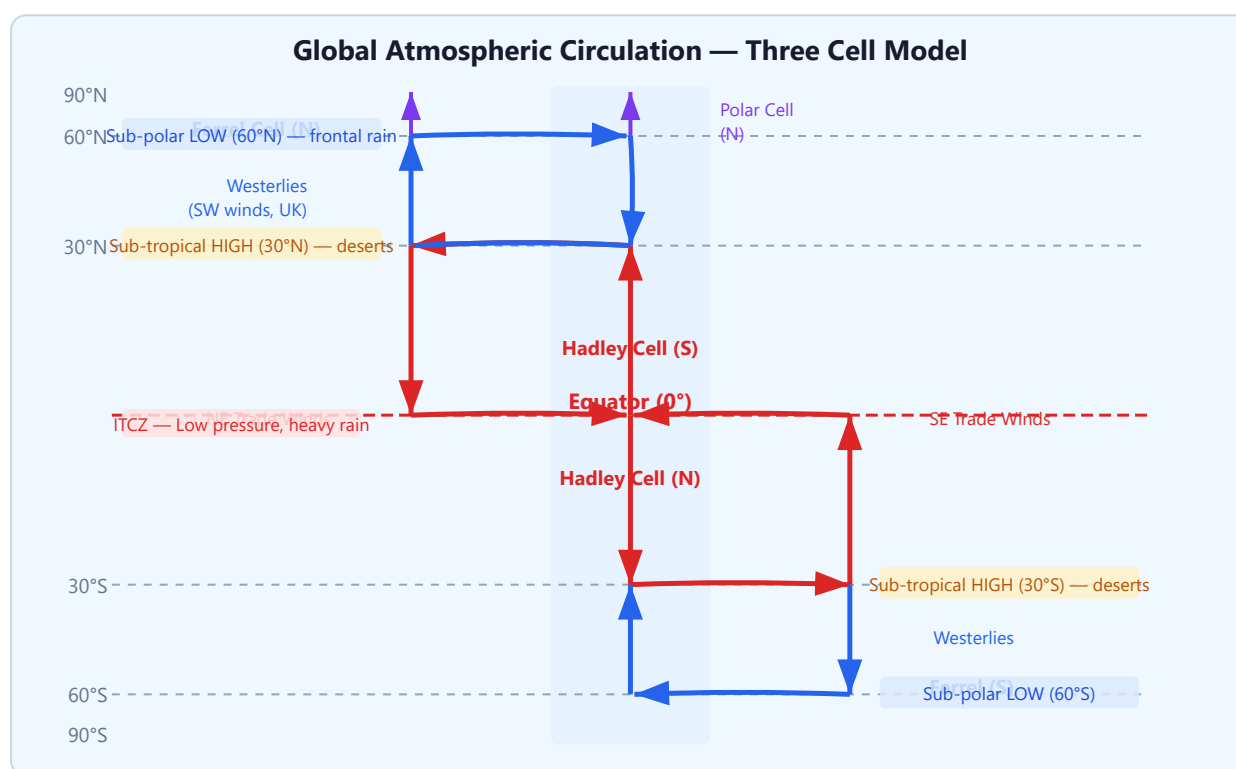


Named example — the Pennines: The Pennine hills run north–south through northern England. Prevailing south–westerly winds bring moist Atlantic air that rises over the Pennines. Manchester (west side) receives c. 850 mm of rain/year; York (east side, in the rain shadow) receives c. 650 mm/year. The difference is even

more pronounced in the higher Lake District fells — Seathwaite in the Lake District averages over 3,000 mm/year (one of England's wettest places), while the Vale of York to the east averages c. 600 mm/year.

4. Global Atmospheric Circulation

Global atmospheric circulation: The large-scale movement of air around the Earth driven by unequal heating. The equator receives more solar radiation than the poles, creating temperature and pressure differences. This drives a system of convection cells (Hadley, Ferrel, and Polar cells) that redistribute heat energy around the globe and create the major wind belts and climate zones.



Pressure Belt / Zone	Latitude	Conditions	Climate result
ITCZ (Inter-Tropical Convergence Zone)	0° (equator)	Rising air, intense heating, low pressure	Equatorial climate: hot, very wet all year; tropical rainforest
Sub-tropical high pressure	~30°N/S	Descending air, high pressure, dry	Hot desert climates (Sahara, Arabian, Australian deserts)

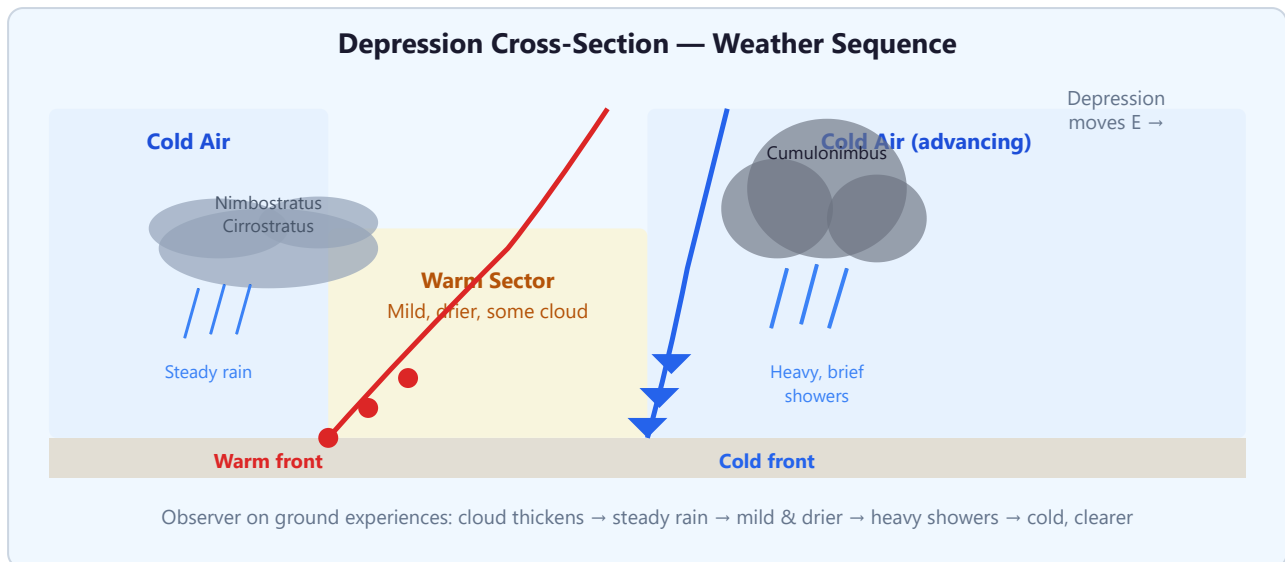
Sub-polar low pressure	~60°N/S	Cold polar air meets warm tropical air; fronts; rising air	Temperate climate with frontal rainfall (UK, NW Europe)
Polar high pressure	90°N/S	Descending cold air, high pressure, very dry	Polar/tundra climate: very cold, low precipitation (mainly snow)

5. Depressions

Depression (mid-latitude cyclone): An area of low atmospheric pressure formed when cold polar air meets warm tropical air. The boundary between these air masses is called the **polar front**. Warm air rises over cold air, creating low pressure at the surface. Depressions bring unsettled, cloudy, wet, and windy weather to the UK. They move broadly west to east across the UK, driven by the jet stream.

Structure of a Depression

Feature	Description
Warm front	Leading edge of the warm air mass, shown on maps as a line with semicircles. Warm air rises gently over cold air ahead of it — produces wide cloud cover and steady, prolonged rain (or drizzle). Passes before the warm sector arrives.
Warm sector	Area of warm air between the warm and cold fronts. Relatively mild, drier, and clearer spells (though some low cloud possible).
Cold front	Trailing edge where cold air undercuts and lifts the warm air rapidly — shown on maps as a line with triangles. Produces brief but intense rainfall, heavy showers, thunderstorms, then cold, showery conditions as the front passes.
Occluded front	As the faster cold front catches up with the warm front, the warm sector is lifted off the ground, forming an occluded front. The depression is weakening at this stage.



Weather sequence as a depression passes over the UK (west to east):

1. **Before warm front:** Cirrus clouds high in sky, then thickening cloud (cirrostratus, altostratus, nimbostratus). Pressure falling. Steady, prolonged rain begins.
2. **In warm sector:** Warmer temperatures. Rain eases. Possible drizzle; low cloud. Wind veers (e.g. from S to SW).
3. **Cold front passes:** Heavy, brief rain or thunderstorms. Winds strengthen and veer to NW. Temperature drops sharply.
4. **After cold front:** Cold, bright, showery conditions. Cumulus/cumulonimbus clouds. Good visibility between showers. Pressure rising.

6. Anticyclones

Anticyclone: An area of **high atmospheric pressure** where air descends from upper levels toward the surface, then diverges outward. Descending air is warmed by compression, which suppresses cloud formation. Anticyclones bring settled, dry, calm weather. They rotate clockwise in the Northern Hemisphere. They can persist for days or weeks, causing blocking weather patterns.

Feature	Summer Anticyclone	Winter Anticyclone
Temperature	Hot, sunny (can reach 30°C+ in SE England)	Very cold (frost common; temperatures below 0°C at night)

Precipitation	Dry; drought risk with prolonged anticyclones	Dry; any moisture forms fog or frost rather than rain
Cloud cover	Clear skies; intense sunshine	Clear at night (radiation fog); possible low stratus by day
Wind	Calm, light winds	Calm; cold temperatures persist because winds don't bring Atlantic warmth
Hazards	Heatwave, drought, UV risk, hosepipe bans	Black ice, freezing fog, hypothermia risk for elderly
Air quality	Poor in urban areas — pollutants trapped (temperature inversions possible)	Smog possible; poor visibility in fog

Depression vs Anticyclone Comparison

Feature	Depression	Anticyclone
Pressure	Low (<1000 mb)	High (>1013 mb)
Air movement	Air converges and rises	Air descends and diverges
Wind speed	Strong; isobars close together	Light or calm; isobars widely spaced
Wind direction (N. Hemisphere)	Anticlockwise (cyclonic)	Clockwise (anticyclonic)
Cloud / precipitation	Heavy cloud; prolonged rain or showers	Clear skies; dry
Duration	Days (pass through quickly)	Days to weeks (blocking high)
Temperature	Variable; warm sector, then colder	Hot in summer; very cold in winter

Exam tip: A common 4-mark question asks you to describe the weather associated with either a depression or anticyclone. Use the table above and refer to temperature, precipitation, cloud cover, and wind. Always specify summer or winter for anticyclones as they produce very different conditions.

7. Extreme Weather Events in the UK

Case Study: Summer Heatwaves

2003 European Heatwave: France and southern UK experienced temperatures 6–7°C above average for over two weeks. UK recorded 38.5°C (Brogdale, Kent). Caused by a persistent anticyclone blocking Atlantic systems. Over 70,000 excess deaths across Europe; 2,000+ in UK. Crops failed; wildfires; power stations shut as river water temperatures rose too high for cooling.

2018 UK Heatwave: Prolonged high-pressure system from June to August. Temperatures reached 35.3°C (Faversham, Kent). UK's joint-hottest summer on record at the time. Drought conditions: rivers at record lows; hosepipe bans; moorland fires in Greater Manchester (Saddleworth Moor).

2022 UK Heatwave: On 19 July 2022, the UK recorded its highest ever temperature: **40.3°C** at Coningsby, Lincolnshire. Caused by a plume of hot air from North Africa channelled north by an anticyclone. Wildfires in London suburbs; rail lines buckled; Luton Airport tarmac melted; schools and businesses closed. Linked to climate change — Met Office stated such temperatures would have been virtually impossible without human-caused climate change.

Case Study: 2013–14 Somerset Levels Flooding

Winter 2013–14 was the UK's wettest in 248 years of records. A series of Atlantic depressions brought relentless storms. The Somerset Levels — a low-lying floodplain in SW England — were inundated from late December 2013 to March 2014.

- 65 sq km of farmland flooded; water depths up to 1 m; 17 million tonnes of water on the Levels
- Villages including Moorland and Muchelney were cut off for weeks by road
- Over 600 homes flooded and 16 farms evacuated; about 1,000 livestock moved to safety
- Somerset County Council initially estimated over £10 million of damage; a later economic impact assessment put the total cost to the county at up to £147.5 million

- Controversy over the Environment Agency's decision to reduce dredging of rivers Tone and Parrett in prior years
- Long-term response: £100 million Somerset Levels Flood Action Plan; 8 km of rivers dredged; upgraded pumping stations; raised flood banks

Exam tip: When answering extreme weather questions, always include: specific dates, specific statistics (temperatures, cost, areas affected), social impacts AND economic impacts, and — for higher marks — link to the cause (e.g. blocking anticyclone, exceptional run of Atlantic depressions).

8. Global Climate Zones

Climate Zone	Location	Temperature	Precipitation	Example
Equatorial	0–5° N/S (equator)	Hot all year: 25–28°C; no seasons	Very heavy: 2,000– 3,000 mm/yr; rain every afternoon	Amazon Basin; Congo Basin; Borneo
Hot semi-arid / tropical savanna	5–20° N/S	Hot all year: 20–32°C; wet season cooler	Strong wet and dry seasons: 750–1,500 mm (seasonal)	Nigeria; Kenya; northern Australia
Arid / Desert	20–30° N/S (sub- tropical)	Extreme range: 40– 50°C day; near 0°C night	Very low: <250 mm/yr; unreliable	Sahara; Arabian Desert; Atacama
Mediterranean	30–40° N/S (W coasts)	Hot dry summers (25– 35°C); mild wet winters (8–15°C)	500–800 mm; mostly in winter	Spain; Italy; California; SW Australia
Temperate (maritime)	40–65° N/S (W coasts)	Mild: 0–22°C; no extremes	Regular rainfall all year: 600–1,500 mm	UK; France; NW USA; New Zealand

Continental	40–65°N (inland)	Extreme range: –20 to +30°C; cold winters	Low–moderate: 300–600 mm; concentrated in summer	Russia; Canada; central USA
Polar / Tundra	>65° N/S	Very cold: –30 to +5°C; short cool summer	Very low: <300 mm/yr (mostly snow)	Arctic; Antarctica; northern Siberia

9. Two Contrasting Extreme Weather Events Outside the UK

Eduqas requires a detailed study of **two contrasting extreme weather events located outside the UK** — one caused by a **dominant, long-lasting high pressure system** and one by an **intense low pressure system**. For each you need causes, impacts on different groups of people, and responses. The UK events in section 7 are useful supporting detail but they do not satisfy this requirement on their own.

High Pressure Event — The Horn of Africa Drought, 2020–2023

Where: Somalia, southern and eastern Ethiopia and northern Kenya, with effects across the wider East African region.

Causes: Persistent high pressure and descending, drying air dominated the region for an unprecedented run of seasons. Five consecutive rainy seasons failed — October–December 2020, March–May 2021, October–December 2021, March–May 2022 and October–December 2022 — with the March–May 2022 season the driest in roughly 70 years. The underlying driver was a prolonged La Niña pattern in the Pacific together with unusually warm sea surface temperatures in the western Indian Ocean, which suppressed the rain-bearing systems the region depends on. Attribution analysis published in 2023 concluded that human-caused climate change made a drought of this severity around 100 times more likely.

Impacts on different groups:

- **Pastoralists** were hit hardest. Around 9.5 million livestock died across the affected countries — roughly 4 million in Ethiopia, over 3 million in Somalia and 2.5 million in Kenya — with livestock losses alone valued at about US\$1.5 billion. For a herding household, the herd is both income and savings, so its loss destroys livelihood and capital at once.
- **Farming households** lost successive harvests, pushing food prices up for everyone, including urban families who do not farm.

- **Women and girls** travelled further for water and fuel, and school attendance fell, with reports of rising early marriage as families sought to reduce the number of dependants.
- **The displaced:** around 1.4 million Somalis were displaced by the drought, moving to camps around towns and cities where services were already stretched. Tens of thousands of people are estimated to have died, mostly children, from the combination of hunger and disease.
- **The region as a whole:** some 46 million people were affected by the drought and the flooding that followed it when the rains finally returned in unusually heavy form.

Responses: Emergency food and cash transfers from the World Food Programme and NGOs; emergency water trucking and borehole repair; livestock offtake schemes that buy weak animals before they die so herders keep some value; therapeutic feeding for malnourished children; and, longer term, drought-tolerant seed distribution, index-based livestock insurance that pays out when satellite data show pasture has failed, and improved early warning through the Famine Early Warning Systems Network. A recurring criticism is that funding arrives after a famine is declared rather than when early warning systems first signal one — a response problem, not a forecasting one.

Low Pressure Event — Typhoon Haiyan, Philippines, November 2013

Where: The central Philippines, striking the islands of Samar and Leyte on 8 November 2013. The city of Tacloban, at the head of a funnel-shaped bay, was worst affected.

Causes: Haiyan formed over the western Pacific where sea surface temperatures were well above 27°C to considerable depth, supplying the latent heat that drives a tropical cyclone. Low wind shear allowed the storm to organise, and it intensified rapidly as it tracked west. At landfall it had maximum sustained winds of around 195 mph (about 315 km/h), among the strongest ever recorded at landfall. The extremely low central pressure raised sea level beneath the storm, and the shallow, funnel-shaped San Pedro Bay amplified the resulting **storm surge** to roughly 5–7 metres at Tacloban.

Impacts on different groups:

- **Coastal residents of Tacloban** bore the great majority of deaths. The Philippine disaster agency recorded about 6,300 killed, with a further 1,062 missing and more than 28,000 injured; around 92% of the deaths in the worst-hit region were on Leyte. Most were caused by the storm surge rather than the wind — many people had evacuated inland from the wind but not upward or far enough from the shore, partly because the term "storm surge" was not widely understood.

- **The poorest households** lost most, because lightweight coastal housing offered no protection and few had savings or insurance.
- **Fishing and farming communities** lost boats, gear and coconut plantations that take years to replace, so the loss of income outlasted the loss of shelter.
- **Survivors in the days afterwards** faced blocked roads, a destroyed airport terminal and no power or clean water, which delayed aid and led to looting and disorder — showing how the secondary effects of an event can rival the primary ones.

Responses:

- **Immediate:** International search and rescue, field hospitals, and large-scale food, water and shelter distribution; the Philippine government declared a state of national calamity.
- **Short term:** Cash-for-work schemes paying residents to clear debris, which restored both the town and household income at the same time.
- **Long term:** A "build back better" reconstruction programme, relocation of some housing out of the highest-risk coastal strip, rebuilding to stronger construction standards, restoration of mangroves as a natural buffer, and reform of the warning system so that it communicates expected surge *height* in metres rather than using unfamiliar technical terms.

Comparing the Two Events

Feature	Horn of Africa drought (high pressure)	Typhoon Haiyan (low pressure)
Onset	Very slow — developed over more than two years	Sudden — the destructive period lasted hours
Spatial extent	Vast, covering several countries	Concentrated along a storm track and a coastline
Main mechanism of harm	Loss of water, pasture and harvests, leading to hunger and displacement	Wind and above all storm surge, causing immediate death and destruction
Warning	Excellent — failing rains were forecast seasons ahead, but funding lagged	Good forecasting of track and strength, but the surge risk was poorly communicated

Recovery	Very slow — rebuilding a herd takes years and a single good rainy season does not restore a livelihood	Visible reconstruction within years, though livelihoods based on coconut and fishing recovered more slowly
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Exam Tip — Why Eduqas Pairs These: The pairing is deliberate. It lets an examiner ask why a slow-onset hazard with excellent warning can still kill tens of thousands while a sudden-onset hazard with a few days' notice kills thousands. The answer in both cases is vulnerability, not the weather: whether people have assets to fall back on, whether the warning is understood, and whether help arrives before or after the damage is done.

Where climate change is covered: In Eduqas Geography B, the evidence for climate change during the Quaternary period, its causes, its consequences, differing attitudes towards it and the responses of individuals and government all sit in **key idea 2.4, Climate Change — Cause and Effect**. They are examined there, not here. What belongs in 2.3 is the *weather*: the characteristics of the UK climate, the global circulation that creates climate zones and pressure systems, and the extreme weather events that result. Use 2.4 for the climate change material, and cross-refer between the two in extended answers — for example when explaining why the frequency and magnitude of drought and tropical cyclones are changing over time.